

TV GETS A FACELIFT

Television is one space that has made a lot of progress in the past year and promises to be a hot space in the coming year too. Exciting features are being added to existing models, while new technologies are invading the flat and high-definition television (HDTV) space. The market for television, as always, is very high. All in all, a high-growth area.

Last year's forecasts spoke about the emergence of HDTV as a key development in the consumer electronics space. This year too, HDTV is a hot topic. Almost all brands now have HDTVs on sale. Service providers such as Tata Sky and Reliance Big TV are offering HDTV content. Increase in paid, interactive services and more channels along with a fall in prices will lead to a greater demand for HDTVs in the coming year.

Plus, the other hot feature cropping up in the HDTV space is Internet convergence. Several models available today, from brands including Sony, Samsung, LG, etc. can be connected to the Internet – enabling the viewer to connect to photo-sites like Flickr or Picasa and view their pictures on a flat screen, stream video from sites such as YouTube, listen to music from the Web, etc.

As we mentioned in our November 2009 article on flat-screens, Internet connectivity could be used by manufacturers for remote diagnosis of problems in your television, delivery of software or driver updates, etc. Television manufacturer Sharp has already started offering such services in the international market.

Another emerging technology that promises greater power for the television is Diiva – the digital interactive interface for video and audio – devel-



oped by a consortium of consumer electronics makers in China. The Diiva 1.0 specification is now available. Diiva refers to an all-in-one cable that combines a high-speed, bi-directional data channel; uncompressed high-definition video; and multichannel audio capability to allow users to connect, configure and control various consumer electronics (CE) devices from their digital TVs, while enjoying an enhanced multimedia experience at the same time.

A recent report by the Consumer Electronics Association, USA, predicts that 2010 might see the rise of 3D television and content, although the market might not grow at such a speedy rate as that of HDTV did. They have based this forecast on surveys conducted by them wherein customers who had seen 3D movies in theatres expressed a keen interest in 3D TV sets and 3D channels. While manufacturers have already started offering 3D capable models, the demand and the number of models available might grow once 3D channels become available. Satellite television providers such as Sky TV in the UK and DirecTV in the US are hinting at the launch of 3D movie channels in the coming year.

SENSORS

Man likes to control everything, and the first step to controlling everything is to know more about whatever he wishes to control. Perhaps it is this basic psychology that has led to the notable growth in the field of sensors – both in terms of technology and the market.

The demand for sensors is riding a high tide now, the process industry is faced with the imperative need for greater quality control, which cannot be managed without monitoring every tiny detail. Sensors are embedded almost everywhere... to measure and monitor the speed, whereabouts and general health of automobiles; to survey environmental conditions; to monitor safety along industrial borders; to measure the temperature and other parameters of chemical processes; and much more.

Several reasons can be attributed to the growing popularity of sensors – improvements in chip technology have resulted in smaller, cheaper yet more accurate sensors that can be embedded anywhere; today's sensors consume less power and even traditional power sources last for a long time; some new-age sensors can even sustain themselves on the energy weaned from wireless signals; plus, developments in alternate energy have led to sensors that can be fuelled by ambient energy such as solar; wireless technologies have made the installation and use of sensors less cumbersome; advancements in software



technology have ensured that the information collected by the sensors is put to best use whether for monitoring processes or for further trend analysis.

According to the Freedonia Research Group, the demand for MEMS-based and imaging sensors will grow the fastest while process variable sensors to monitor pressure, temperature, flow and level will continue to be the largest segment. The automotive industry will also remain a huge market.

According to the group, till 2012 imaging and proximity or position sensors are expected to record the fastest growth. Imaging sensors – including charge-coupled devices (CCD), complementary metal-oxide semiconductors (CMOS) and thermal (infrared) sensors – will register the fastest gains of all sensor types. Of these, the CMOS-based imaging sensors will be more popular because of their lower costs, lower power requirements and higher speeds.

Thermal imaging sensors will be used more for military applications, by police departments, fire departments, structural testing companies, original equipment manufacturers (OEMs) and private consumers.

The popularity of proximity and positioning sensors will grow fast, owing to increasing applications in automobiles and industrial sector. While the market for camshaft and crankshaft positioning sensors will continue to grow, there will also be a spurt in newer applications such as collision avoidance systems.